

Fet 45

An ultra-fast FET peak limiter: fixed threshold, INPUT drive, four ratios plus ALL, seven-step attack and release, and parallel MIX.



Overview

Fet 45 (from Feldeffekt, field effect) is a peak limiting amplifier in the spirit of the classic ultra-fast FET limiters of the late 60s and 70s, dressed as mid-80s state radio and TV transmission gear. It is the fourth of the family with Pegel 45, Stimm 45 and Raum 45.

It works the way the original does. The threshold is fixed, so you do not set one: INPUT drives the signal harder into the FET, which gives more limiting, and OUTPUT sets the level afterwards. The detector listens to the FET's output (feedback style), which gives the classic soft, curved knee. ATTACK and RELEASE are numbered 1 to 7, with 7 the fastest, and the attack goes down to 20 microseconds - fast enough to catch almost any transient, or to let a little through for punch.

Five ratio keys choose 4, 8, 12 or 20:1, or ALL, the famous "all buttons in" mode: an aggressive, pumping, distorted squash where transients punch through and everything else pumps. The FET adds a little second-harmonic colour as it works, more on the low end at slow release settings, and a growl on low notes at the fastest releases; the line amp and output transformer add their own gentle weight when pushed. MIX blends in the dry signal for parallel compression.

Fet 45 runs at twice the sample rate inside, so the fast gain changes stay clean. Its latency is 23 samples, reported to REAPER.

Reach for it for

- Grabbing a vocal and pushing it forward
- Evening out bass and adding sustain to guitar
- Snare crack and kick punch with slower attacks
- Room-mic and drum-bus smash with ALL
- Parallel compression with MIX
- Fast, transparent peak limiting at 20:1

Controls

Drag the knobs up and down (hold Shift for fine moves); double-click to reset one; the mouse wheel works too. A knob shows its value while you touch it.

The header

PRESETS	Opens the preset list (or click the display). The list ends with Reset all to defaults, which also resets the theme and the output stage.
◀ / ▶	Step through the presets.
Preset display	Shows the preset number and name. Click it for the list.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

INPUT

dB	The gain into the FET, -24 to +24 dB in half-dB steps (0 dB = unity, the default). The threshold sits at 0 VU (-18 dBFS peak), so the more you turn this up, the more Fet 45 limits.
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TIME

Both knobs are numbered 1 to 7, with 7 the fastest, as on the original.

ATTACK	1 - 7 (3 by default): 800 microseconds down to 20 microseconds. Slow settings let the front of each hit through, for punch; fast settings catch everything.
RELEASE	1 - 7 (7, the fastest, by default): 1.1 s down to 50 ms. The release depends on the programme: short peaks recover in the set time, but after a long passage of limiting the last part comes back more slowly, which keeps sustained material from pumping.

Meter

The meter in the middle shows what the METER keys choose. In the ENAMEL theme it is a needle meter (gain reduction is read on the printed VU scale, as on the hardware: the needle rests at 0 and swings left). In VFD and MATRIX it is a ladder of 1 dB blocks (gain reduction 0 to -20 dB, lit from the right) with a numeric readout.

OUTPUT

dB	The level after the FET, -24 to +24 dB (0 dB by default). Use it to make up for the limiting.
MIX	Parallel compression: the blend of the limited signal with the dry input, 0 - 100% (100% = all FET, the default). The dry path is delayed to line up exactly.
DRIFT	0 - 100% (50% by default). See Drift below.

RATIO

4 / 8 / 12 / 20	The ratio above the knee (4:1 by default). The knee is soft and curved, wider at the lower ratios (about 8 dB at 4:1, 4 dB at 20:1). 20:1 is a true peak limiter.
ALL	"All buttons in": about 40:1 just over the threshold, bending to about 12:1 higher up, with a lazier attack, a detector that overshoots and digs in after every hit then lets go

(pumping), a quicker release and more harmonic distortion. Transients punch through and everything else pumps.

METER

GR / OUT / IN / OFF

What the meter shows: gain reduction (the default), the output level, the input level (before INPUT), or nothing. OUT and IN read in VU, with 0 VU = a sine peaking at -18 dBFS, which is also where the threshold sits (THR = 0 VU).

THEME

ENAMEL

Bone-grey baked enamel with knurled aluminium knobs and a needle meter (the default).

VFD

Charcoal slate with a cold turquoise glow, dark bakelite knobs with a white line, and a ladder meter.

MATRIX

Battleship grey with LED blocks and a ladder meter.

LINK

Stereo link, on by default: one detector hears both channels (the louder one), so the image stays steady. Off, the two channels limit on their own.

OFF / CLIP / LIM

The GFX output stage: nothing, a soft clipper just under 0 dB (the default; the CLIP key flashes red while it works), or a look-ahead limiter (it adds 3 ms of latency, compensated by REAPER).

The theme changes only the look, and is saved with the project.

Drift

DRIFT is the wander of the FET's bias. The threshold drifts by up to about half a dB, the release timing by up to 8%, and the two channels track the gain reduction slightly differently; with LINK off, each channel's threshold also wanders on its own. At 0% it is perfectly steady.

Presets

Presets set the ratio, INPUT, OUTPUT, ATTACK, RELEASE, METER, LINK, MIX and DRIFT. The theme and the output stage are left as they are.

Init	4:1, unity in and out, attack 3, release 7.
Vocal Grab	4:1, +8 dB in, -4 dB out, attack 4, release 6: a forward, controlled vocal.
Bass Even	8:1, +6 dB in, slow attack 2 and slow release 2.5 for a smooth, even bass.
Snare Crack	4:1, +10 dB in, the slowest attack so the crack comes through, fast release.
Room Smash (ALL)	ALL, +14 dB in, -9 dB out: the classic pumping room sound.
Drum Bus Parallel	ALL, +12 dB in, MIX 35%: smashed drums blended under the dry ones.
Guitar Sustain	8:1, +12 dB in, attack 5, release 3: long, even sustain.
Broadcast Limit 20	20:1, +4 dB in, fastest attack, the meter on OUT: transparent peak control.
Ost Rundfunk	12:1, +8 dB in, fast attack and medium release, with lots of drift: an old radio-chain sound.
Tokio Vocal	4:1, +10 dB in, attack 5, release 6, little drift: a tight modern vocal.
Kick Punch	4:1, +8 dB in, the slowest attack, MIX 80%.
Gentle 4	4:1, -4 dB in, slower release: a light touch of levelling.

Tips

- Set OUTPUT first to a few dB below INPUT, then turn INPUT up until the GR meter shows the amount of limiting you want, and trim OUTPUT to match the loudness.

- For punch, use a slow ATTACK (1 - 3) so the transient gets through before the FET clamps down.
- ALL on a room mic or drum bus, with MIX at 25 - 50%, gives excitement without losing the dynamics.
- The fastest RELEASE settings make low notes growl; slow the release down on bass if you hear distortion you do not want.
- Switch the METER to IN to see the level arriving (before INPUT): with INPUT at 0 dB, the limiting starts where the loud parts pass 0 VU, and every dB of INPUT moves that point down by a dB.

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